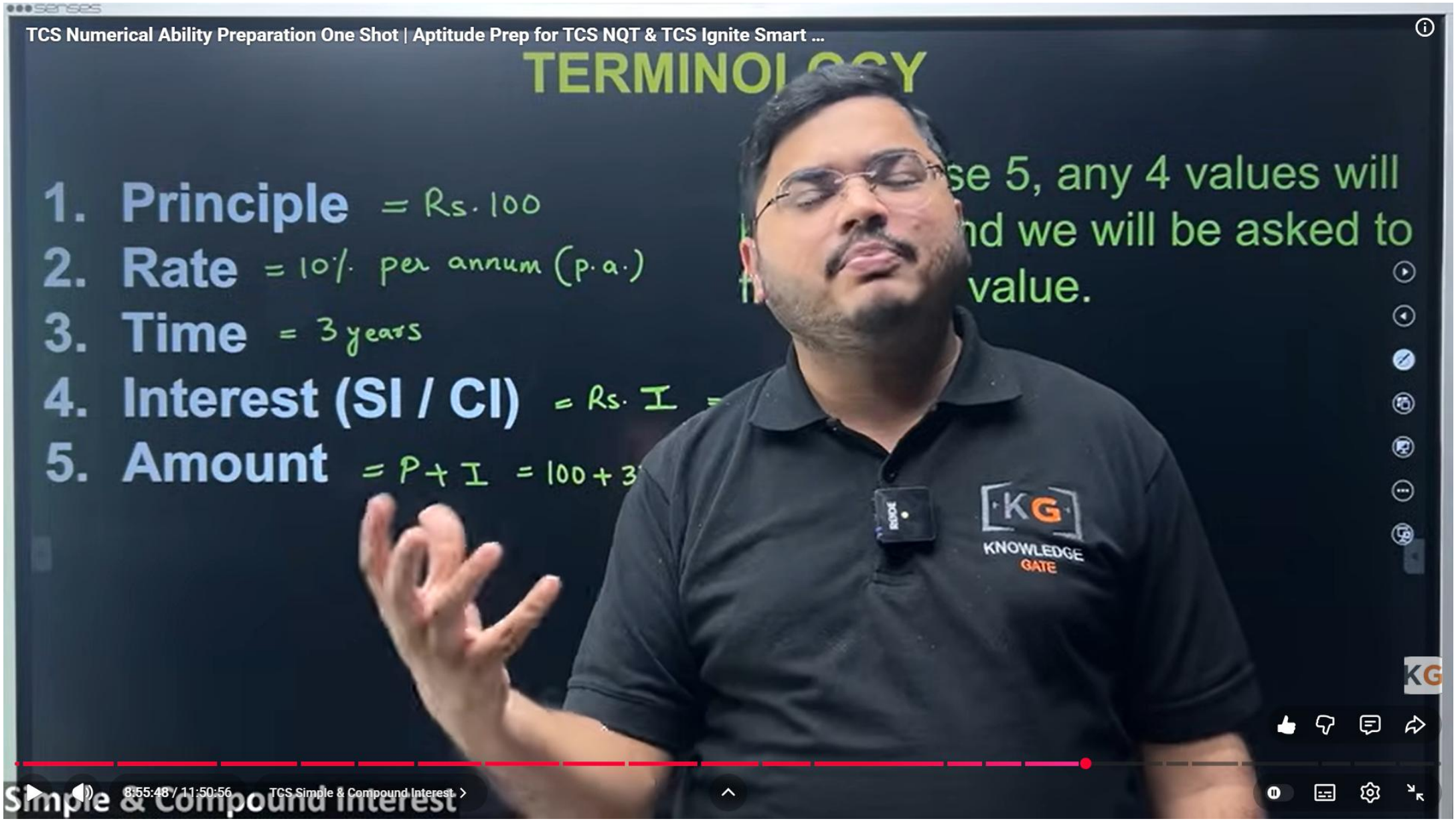

TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
TERMINOLOGY
1. Principle = Rs. 100
2. Rate = 10% per annum (p.a.)
3. Time = 3 years
4. Interest (SI / CI) = Rs. I
5. Amount = P + I = 100 + 3
se 5, any 4 values will
nd we will be asked to
value.
KG
KNOWLEDGE
GATE
8:55:48 / 11:50:56
TCS Simple & Compound Interest
Simple & Compound Interest

TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...

SIMPLE INTEREST & COMPOUND INTEREST

$P = 100$
$R = 10\%$
$T = 3$ years

SI

$100 \xrightarrow{10\%} 10$

$100 \xrightarrow{10\%} 10$

$100 \xrightarrow{10\%} 10$

$10 + 10 + 10 = 30$

$SI = Rs.30$

$A = P + I = 100 + 30 = Rs.130$

CI

$100 \xrightarrow{10\%} 10$

At the end of 1 year: $100 + 10 = 110$

$110 \xrightarrow{10\%} 11$

At the end of 2 years: $110 + 11 = 121$

$121 \xrightarrow{10\%} 12.1$

At the end of 3 years: $121 + 12.1 = 133.1$

$133.1 \rightarrow 100, 33.1$

$A = 133.1$

$I = 33.1$

8:57:40 / 11:50:56 TCS Simple & Compound Interest
Simple & Compound Interest

TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...

Q1. Find the amount & simple interest on Rs. 6000 at the rate of 6% per annum for 2 years?

M1

$P = 6000$

$R = 6\%$

$T = 2$

$\Big\} \; 12\%$

$SI = \frac{P \times R \times T}{100}$

$= \frac{6000 \times 6 \times 2}{100}$

$= 60 \times 12$

$SI = \text{Rs. } 720$

$A = 6000 + 720 = \text{Rs. } 6720$

M2

$6\% \rightarrow 1$ year

$12\% \rightarrow 2$ years

Total interest $= 12\%$

$(RT)\%$

$SI = 12\%$ of 6000

$\frac{12}{100} \times 6000$

12×60

$= 720$

10% 1% 1%

600 60 60

720

Share

9:07:16 / 11:50:56 TCS Simple & Compound Interest

Simple & Compound Interest

TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...

Q3. Find the Simple Interest on Rs. 2500 at the rate of 5% per annum for 219 days.

$\times \frac{3}{75}$

5% → 1 year (365 days)

1% 1% 1% 1% 1%

3%

73 73 73 73 73

219 days

$\frac{365}{5} = 73$

365 days → 5%

1 day → $\frac{5}{365}$ %

219 days → $\frac{5}{365} \times 219 = 3\%$

Total Interest = 3% ⇒ 3% of 2500

1% 1% 1%

25 25 25

75

$\boxed{SI = Rs.\ 75}$

A = 2500 + 75 = 2575

9:11:27 / 11:50:56 TCS Simple & Compound Interest
Simple & Compound Interest

TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...

Q4. What will be the simple interest on Rs.3200 at the rate of 5% per annum from 4th April to 16th June.

Interest will start from 5th April to 16th June

5 April — 30 April = 26 days

1 May — 31 May = 31 days

1 June — 16 June = 16 days

26+31+16 = 73 days

SI = Rs. 32

A = 3200+32 = 3232

365 × 1/5th = 73

5% × 1/5th 1%

5% → 365 days

1% 1% 1% 1% 1%

73 73 73 73 73

1% = 73 days

Total Interest = 1%

⇒ 1% o

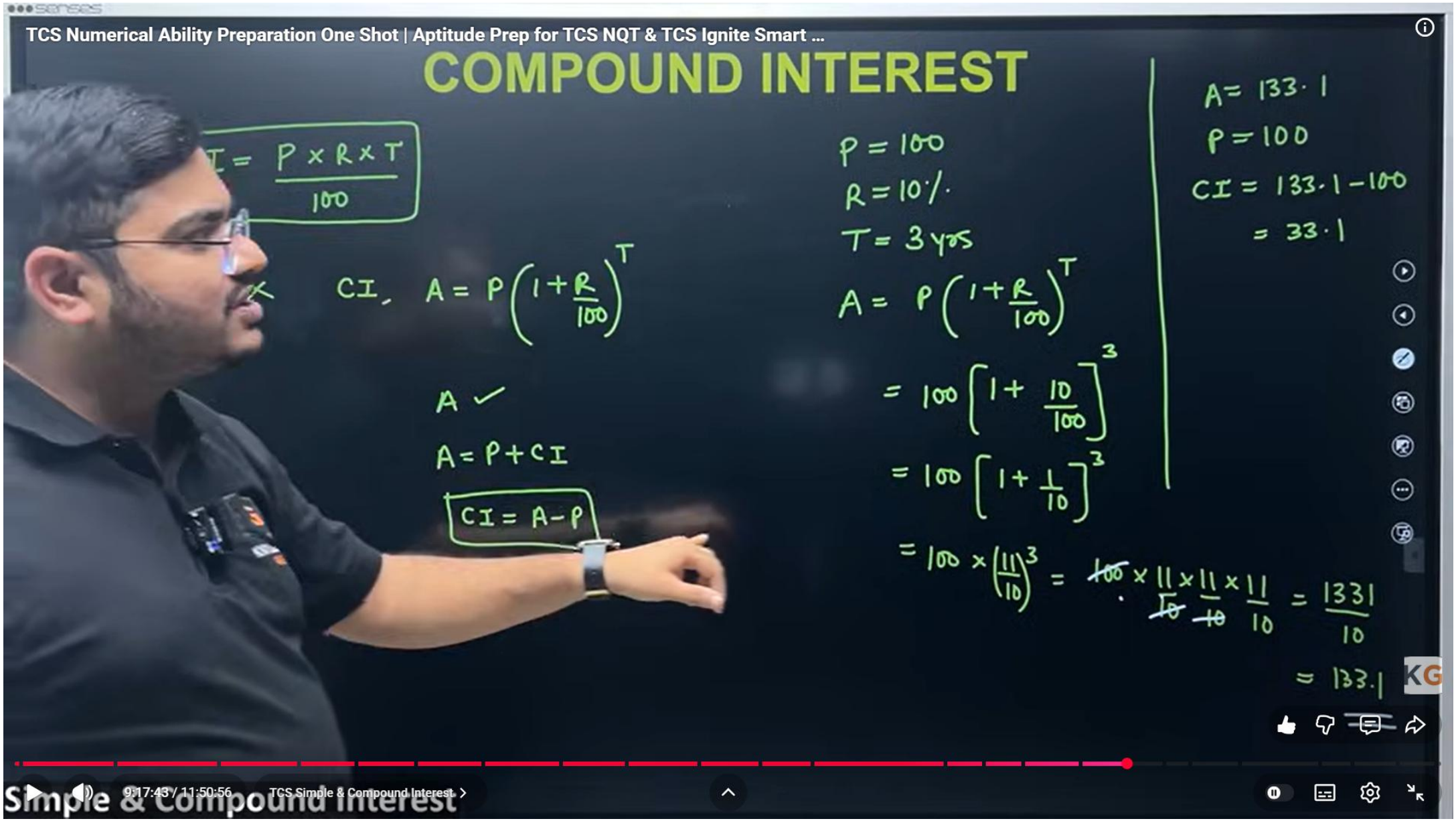
= Rs

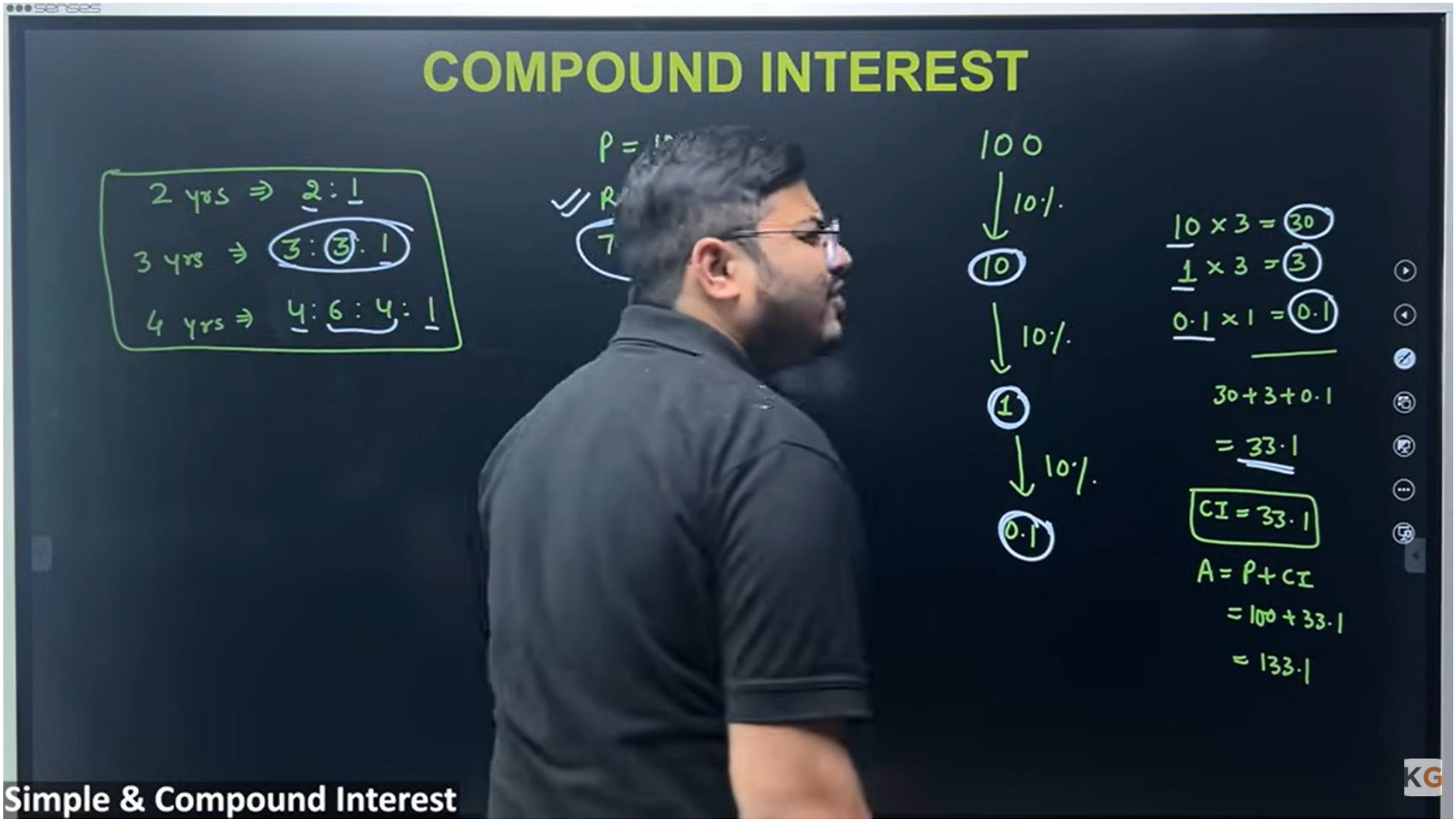
TATA CONSULTANCY SERVICES

Share

9:14:21 / 11:50:56 TCS Simple & Compound Interest

Simple & Compound Interest


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
COMPOUND INTEREST
I = P × R × T / 100
CI, A = P(1 + R/100)^T
A ✓
A = P + CI
CI = A − P
P = 100
R = 10%
T = 3 yrs
A = P(1 + R/100)^T
= 100[1 + 10/100]^3
= 100[1 + 1/10]^3
= 100 × (11/10)^3 = 100 × 11/10 × 11/10 × 11/10 = 1331/10
= 133.1
A = 133.1
P = 100
CI = 133.1 − 100
= 33.1
9:17:43 / 11:50:56
TCS Simple & Compound Interest
Simple & Compound Interest


COMPOUND INTEREST
2 yrs ⇒ 2 : 1
3 yrs ⇒ 3 : 3 : 1
4 yrs ⇒ 4 : 6 : 4 : 1
100
10%
10
10%
1
10%
0.1
10 × 3 = 30
1 × 3 = 3
0.1 × 1 = 0.1
30 + 3 + 0.1
= 33.1
CI = 33.1
A = P + CI
= 100 + 33.1
= 133.1
Simple & Compound Interest

TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...

Q6. What will be the compound interest on Rs. 45000 at the rate of 10% per annum for 2 years?

2 years

2:1

45000

10%

4500

10%

450

4500 × 2 = 9000

450 × 1 = 450

CI = 9000 + 450

= 9450

6550

Answer cross check

10% ← 1 yr

20% ← 2 yr

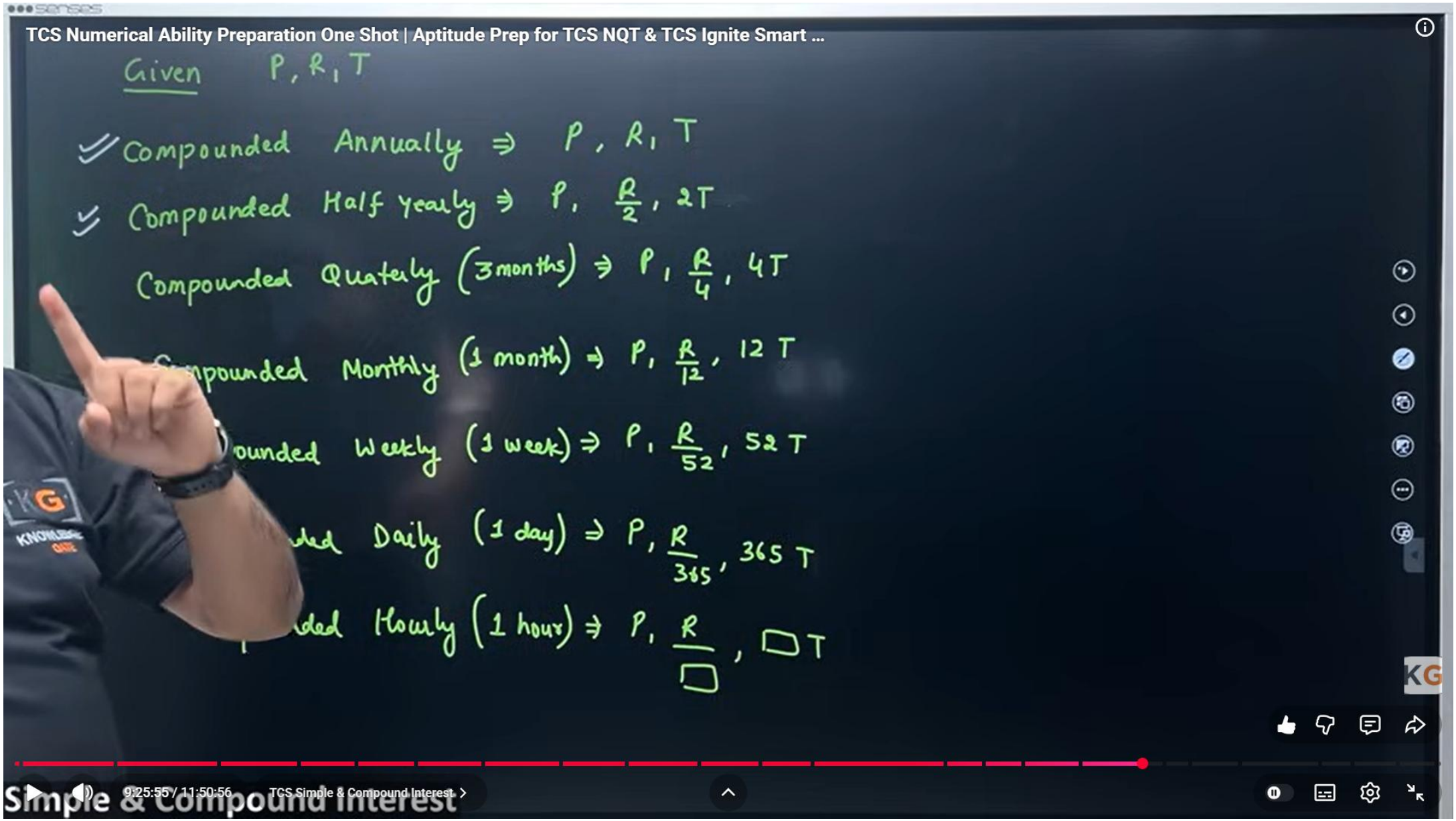
20% of 45000

SI = 9000

CI > SI

9:22:32 / 11:50:56 TCS Simple & Compound Interest

Simple & Compound Interest


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
Given P, R, T
Compounded Annually ⇒ P, R, T
Compounded Half yearly ⇒ P, R/2, 2T
Compounded Quately (3 months) ⇒ P, R/4, 4T
Compounded Monthly (1 month) ⇒ P, R/12, 12 T
Weekly (1 week) ⇒ P, R/52, 52 T
Daily (1 day) ⇒ P, R/365, 365 T
Hourly (1 hour) ⇒ P, R/□, □T
9:25:55 / 11:50:56
TCS Simple & Compound Interest
Simple & Compound Interest

TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...

Q9: What will be the compound interest on Rs. 16000 at the rate of 20% annually for 9 months if the interest is compounded quarterly.

P = 16,000
R = 20%
T = 9 months

P = 16,000
R = 20/4 = 5%
T = 4 × 9 = 36 months
= 3 yrs

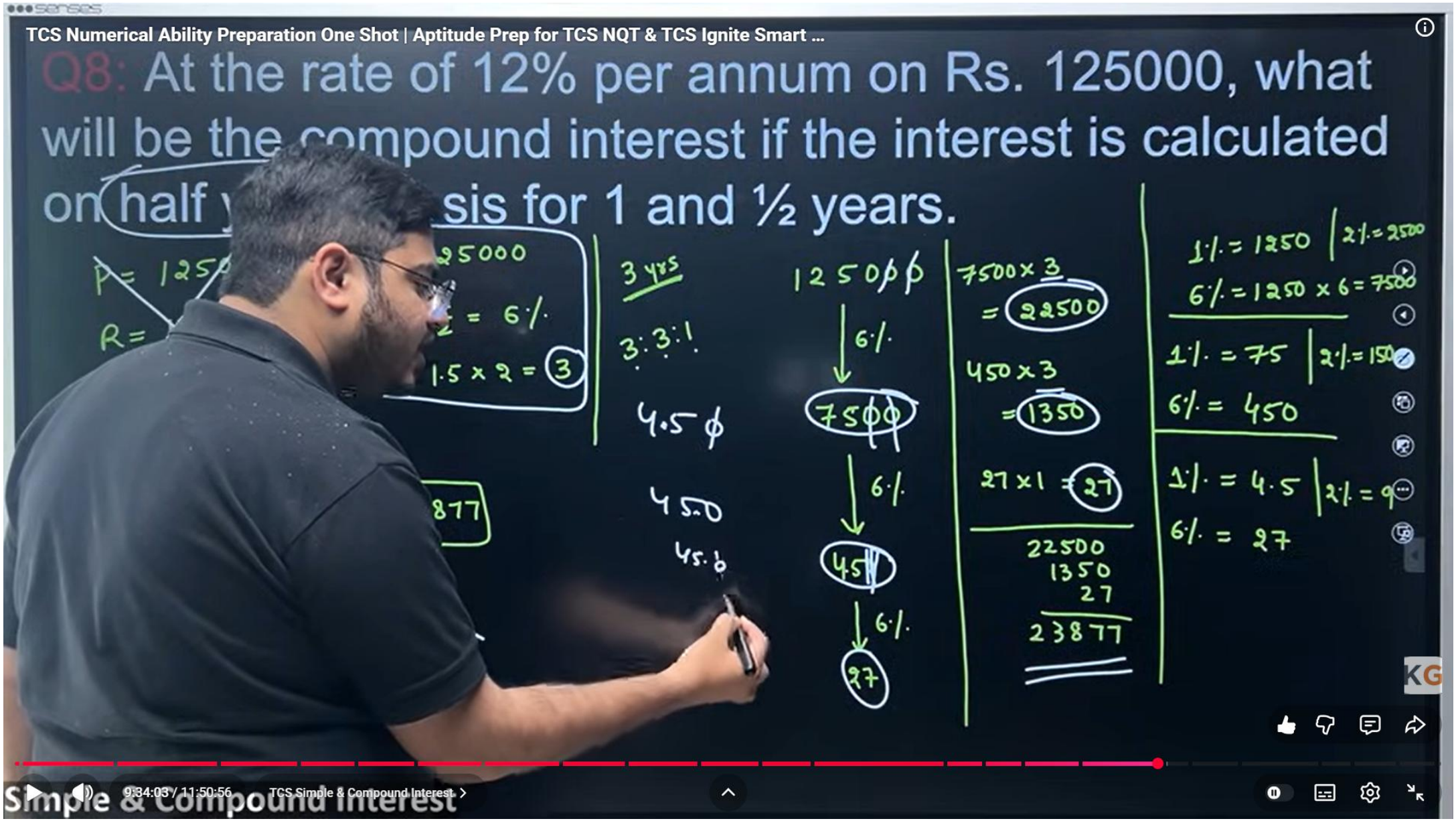
3 yrs
3 : 3 : 1

16000
↓ 5%
800
↓ 5%
40
↓ 5%
2

800 × 3 = 2400
40 × 3 = 120
2 × 1 = 2
2522

CI = Rs. 2522

9:34:56 / 11:50:56 TCS Simple & Compound Interest
Simple & Compound Interest


TCS Numerical Ability Preparation One Shot | Aptitude Prep for TCS NQT & TCS Ignite Smart ...
Q8: At the rate of 12% per annum on Rs. 125000, what will be the compound interest if the interest is calculated on half y sis for 1 and ½ years.
25000
= 6%
1.5 x 2 = 3
877
3 yrs
3:3:1
4.5
45.0
45.6
12500
6%
7500
6%
450
6%
27
7500 x 3
= 22500
450 x 3
= 1350
27 x 1 = 27
22500
1350
27
23877
1% = 1250
2% = 2500
6% = 1250 x 6 = 7500
1% = 75
2% = 150
6% = 450
1% = 4.5
2% = 9
6% = 27
9:34:03 / 11:50:56
TCS Simple & Compound Interest
Simple & Compound Interest